Lesson Planning

Complete Lesson Plan Book

Class: FSc Pre-Engineering · Class 11

Subject: Physics

Medium: English

Academic Year 2026-27

DM

This book is prepared as a teaching guide for teachers.

Table of Contents

Unit 1: Newton's Second Law of Motion: F = ma

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Learning Objectives

• Students will state Newton's second law of motion in words and in the form F = ma.

• Students will calculate force, mass or acceleration when the other two quantities are given.

• Students will explain everyday examples of the law, such as pushing a car and kicking a football.

Key Concepts

• Newton's second law says the acceleration of an object is directly proportional to the net force on it and inversely proportional to its mass.

• The formula is F = ma, where force is measured in newtons, mass in kilograms and acceleration in metres per second squared.

• A net force causes acceleration in the direction of the force; balanced forces cause no acceleration.

• Doubling the force doubles the acceleration, while doubling the mass halves the acceleration for the same force.

Teaching Methodology (Step by Step)

1. Ask why a small push moves an empty trolley easily but barely moves a full one, and collect students' ideas.

2. State Newton's second law and write F = ma on the board, explaining each symbol and its unit.

3. Work out two examples together: finding force from mass and acceleration, then finding acceleration from force and mass.

4. Students solve three problems in pairs, checking units before writing the answer.

5. Students explain one everyday example of the law in their own words.

Classroom Activities

• Force race: students predict which push moves a toy car further, then test it.

• Mass test: the same push is applied to an empty box and a box full of books, and results are compared.

• F=ma cards: students match a force, mass and acceleration card set that fits the formula.

Assessment Questions

1. Write Newton's second law in words and as F = ma.

2. A force of 10 N acts on a mass of 2 kg. What is its acceleration?

3. Why does a heavier trolley need a bigger push to reach the same speed as a lighter one?

Homework

• A 5 kg object accelerates at 3 m/s². Find the net force acting on it.

• Explain in four lines why it is harder to push a loaded cart than an empty one, using F = ma.

Time Allocation

40 minutes